Jacob Ogle

Ottawa Lake, MI — (419) 202-2012 — jakeogle@umich.edu
linkedin.com/in/jakeogle — github.com/jake-ogle

Summary

Software engineer specializing in ML infrastructure and high-performance computing, with experience building petabyte-scale data workflows, GPU platforms, and production data systems across automotive, manufacturing, and clinical research.

Experience

Senior Machine Learning Engineer Jun 2026 – Present
Ford Motor Company – ADAS DataOps *Dearborn, MI*

- Designed and implemented a CLI- and Slurm-driven workflow for parallel MPI scanning, batched tar creation, and GCS archival across ADAS filesystems holding over 20 PB, supporting archival operations of up to 1 PB.
- Built filesystem metadata parsing pipelines and deployed Grafana and PostgreSQL on Kubernetes to monitor Lustre and GPFS capacity and quantify cold-data archival candidates using a 365-day modification-time threshold.
- Deploy and operate Voxel51 ML tooling on Kubernetes with GPU access, offloading compute workloads to Slurm batch systems.
- Build cloud pipelines to process ML datasets using open-weight models, Apache Airflow for orchestration, and Apache Beam on Google Cloud Dataflow.

Senior Machine Learning Engineer Dec 2025 – May 2026
Michigan Medicine – Machine Learning in Neurosurgery (MLiNS) Lab *Ann Arbor, MI*

- Owned HPC data infrastructure for multimodal clinical AI research, expanding a 500 TB dataset to near-petabyte scale through public dataset integration and clinical data acquisition.
- Built HPC ingestion and preprocessing pipelines using dcm2niix and custom C++ metadata parsers, generating study-level configuration files for MRI, CT, and pathology workflows.
- Managed Slurm workflows and GPU allocations for six PhD researchers, allocating eight-GPU nodes per training job and supporting single-node LLM serving across eight GPUs.
- Built Apptainer/Singularity containers and Slurm launch scripts for single-node, multi-GPU training of vision and vision-language models; configured torch.distributed and profiled GPU workloads using NVIDIA tools.
- Developed an MPI-based text extraction pipeline for 600,000–800,000 clinical reports retrieved through Solr NLP searches, distributing HTML reports across worker ranks to extract tagged content for LLM summarization.
- Deployed and operated self-hosted vLLM inference on NVIDIA L40 GPU nodes, integrating Apache Solr and SQL Server for clinical text summarization, annotation, and classification.
- Maintained a research data catalog using DuckLake, DuckDB, SQLite, and SQL Server to organize imaging, pathology, and clinical reports across HPC workflows.

Software Engineer Jan 2023 – Nov 2025
Magna International *Northwood, OH*

- Served as the primary engineer for data infrastructure, internal applications, and applied AI across multiple manufacturing plants with centralized IT infrastructure.
- Provisioned and operated Linux, Docker, and Kubernetes infrastructure for data pipelines, ML applications, and real-time monitoring across molding, painting, assembly, and inspection.
- Designed and built a telemetry platform for 20+ molding presses and assembly and painting robots across three plants, using Go for MQTT ingestion and Kubernetes-hosted TimescaleDB (PostgreSQL) for time-series storage.
- Developed React/TypeScript applications serving 100–150 plant users, including near-real-time production dashboards and HR tools; built a Go/RabbitMQ pipeline delivering out-of-range process alerts to management via Teams.
- Engineered Apache Airflow ingestion and validation pipelines for manufacturing telemetry and enterprise data, integrating on-premises systems with Azure and AWS.
- Contributed to multiple computer vision proofs of concept using OpenCV and YOLO for automated inspection and defect detection across manufacturing processes.

Software Engineering Intern – XC Division May 2021 – May 2022; Jul – Dec 2022
Bosch North America *Plymouth, MI*

- Developed and maintained OTA reflashing software for GM's central control gateway, implementing features and resolving defects in MISRA-compliant C on QNX RTOS.
- Used Docker-based cross-compilation environments to build embedded software for the target platform.
- Built Python automation for software builds and cryptographic code signing, implementing GM key management requirements for OTA reflashing.
- Contributed to CI/CD workflows supporting automated builds, software integration, and delivery of gateway firmware.

Product Development Intern – Vehicle Health Alerts May 2022 – Jul 2022
Ford Motor Company *Dearborn, MI*

- Investigated field-reported vehicle issues with the Vehicle Health Alerts team, including tire pressure monitoring anomalies and ECU faults.
- Queried vehicle diagnostic snapshots using Google BigQuery to identify patterns, document findings, and support engineering investigations.
- Developed hands-on knowledge of vehicle diagnostic data flows, ECU functionality, and connected vehicle systems to interpret reported issues and collaborate with software and systems engineers.

Clinical Data Manager / Research Engineer Jun 2019 – May 2021
Michigan Medicine – Bielas & Martin Labs *Ann Arbor, MI*

- Supported HPC-based computational genomics workflows for the Gabriella Miller Kids First research initiative, working with large-scale whole-genome and whole-exome sequencing datasets.
- Used University of Michigan ARC HPC storage and Globus to organize, manage, and transfer clinical and genomic datasets across research computing environments.
- Developed Python automation for research data processing, metadata management, and validation to support clinical and genomic workflows on HPC infrastructure.
- Built participant tracking databases and data management tools for multi-institutional rare disease research, supporting structured clinical data collection and research coordination.

Education

University of Texas at Austin 2026 – Present
M.S. Computer Science, in progress

Oregon State University 2021 – 2025
Postbaccalaureate coursework in Computer Science

- Selected coursework: Parallel Systems in C++, Algorithms, Open Source Software Development

University of Michigan – Ann Arbor 2016 – 2019
B.S. Chemistry

Technical Skills

Languages: Python, C/C++, Go, Bash, SQL, Rust
Systems & HPC: Linux, QNX RTOS, Slurm, MPI, Apptainer/Singularity, multi-GPU computing, MISRA C
Containers & Infrastructure: Docker, Kubernetes, CI/CD, cross-compilation
GPU & ML: GPU administration, MIG, time slicing, distributed PyTorch training, vLLM, OpenCV, YOLO
Storage & Observability: Lustre, IBM Spectrum Scale (GPFS), Globus, Grafana
Data Platforms: Apache Airflow, Apache Beam, Google Cloud Dataflow, RabbitMQ, MQTT, Apache Solr
Databases & Cloud: DuckDB, DuckLake, SQL Server, PostgreSQL, SQLite; GCP, Azure, AWS

Publications

- Health system learning enables generalist neuroimaging models. *Nature Medicine*, 2026. Coauthored.
- Genotype-phenotype analysis of 523 patients by genetics evaluation and clinical exome sequencing. *Pediatric Research*, 2020. Coauthored.